

PREDICTIVE DISPATCH ENGINE

ALTAIR AI STUDIO PROJECT

Predictive Railway Delay & Cascading Impact Rescheduling System

AI Studio-Based Delay Prediction & Recommendation for a Rail Corridor Network

FEEDER TRAIN

T024 · Suburban

+16.9 min at Kingsford

CONNECTING TRAIN

T031 · Passenger

Transfer buffer at risk



A single upstream delay reaches a second train through a shared connection — this is the pattern the model is built to catch.

PROJECT

Predictive Railway Delay Management

PRESENTER

Aravind

INSTITUTION

Easwari Engineering College

PROGRAM

SIRP – Final Project

Problem Statement & Why This Matters



Network Scale

A 12-station corridor carrying 28 daily services accumulates many small delays across a shared network of track and platforms.



Manual Overhead

Rail traffic controllers currently react to each delay individually, using fixed recovery rules rather than anticipating disruption.



Cascading Effect

A delay at one station rarely stays contained — it propagates through shared platforms and connecting trains to the rest of the network.



Operational Value

Anticipating disruption before it cascades protects connections, platform capacity, and passenger experience network-wide.



CASCADE RISK · TYPICAL PATTERN OBSERVED IN THE DATA

Freight train T-series delayed 18 min at a junction platform

→ connecting Suburban service breaches its transfer buffer, delay reaches a second train within minutes.

Goal: anticipate this chain before it forms, not react after it lands.

Project Objective

MAIN OBJECTIVE

Build an AI Studio-based predictive system that forecasts train delays, models how they cascade to connected trains, and recommends dispatch actions to minimize total network disruption.



01

Analyze Data

Examine 90 days of corridor schedule, delay, and platform data for patterns.



02

Predict Delay Risk

Estimate the probability a train will be delayed at its next stop.



03

Estimate Duration

For at-risk trains, predict how many minutes the delay will run.



04

Model Downstream Impact

Trace how a feeder train's delay affects connected trains at shared stations.



05

Recommend Action

Turn predictions into hold, reassign-platform, or monitor decisions.



06

Present Dashboard

Surface the at-risk watch list for controllers in a simple live view.

Dataset & Attributes

16,380 train-stop events across a 90-day simulated corridor were used, with attributes covering schedule, operational, and environmental context.



ATTRIBUTES USED FOR MATCHING

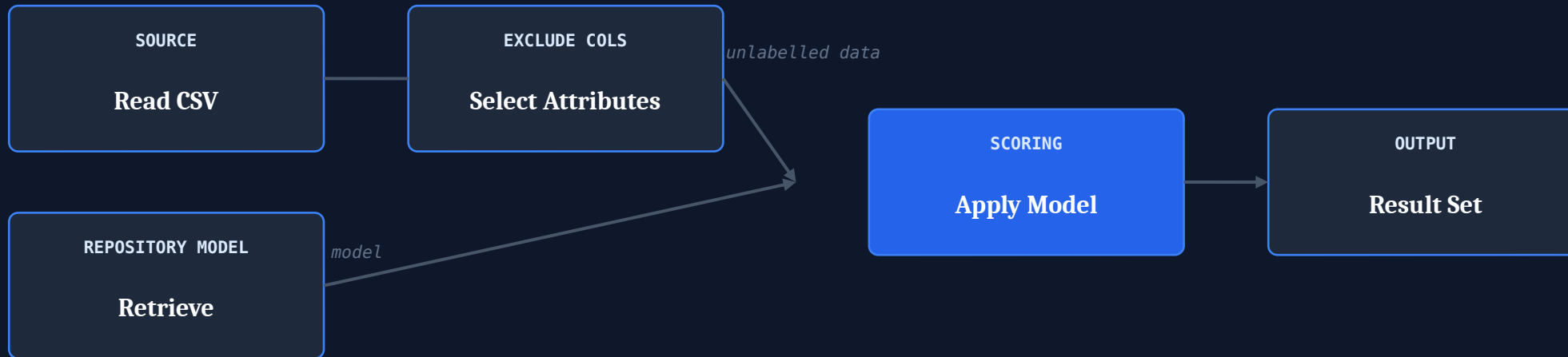
train_type / station_id	Categorical context for each stop
weather / day_of_week	Environmental & calendar context
stop_sequence, is_origin/destination	Position within the train's route
num_platforms, km_from_origin	Station capacity & corridor position
sched_hour, is_rush_hour, is_weekend	Time-of-day congestion signal
prev_stop_delay	Real-time upstream delay — top predictor

EXCLUDED FROM MODELING

- train_id (identifier only)
- date (used for joins, not as a feature)
- delay_severity (derived label, post-hoc)

Feature reduction preserves performance: identifiers and derived labels are dropped so the model learns from operational signal, not row bookkeeping.

Railway_Delay_Prediction_and_Analysis_System.rmp



WORKFLOW SUMMARY

This is the actual submitted process: it reads new schedule data, drops identifier/leakage columns, retrieves the pre-trained “Railway Delay Final Random Forest” model from the AI Studio repository, and applies it to score every row with a delay prediction and confidence.

Delay Probability & the Signal Behind It

Random Forest Classifier

Combines 100–300 decision trees over schedule, weather, and — most importantly — the delay already observed at the train's previous stop, to output a 0–100% risk score for its next arrival.

LOW-RISK STOP

prev_stop_delay = 0 min

12%



HIGH-RISK STOP

prev_stop_delay = 22 min

81%

RISK CLASSIFICATION BANDS

● 60 – 100% High risk — active intervention

● 35 – 59% Medium risk — monitor closely

● 0 – 34% Low risk — no action needed

Screening threshold: 50% — the point above which a stop is treated as "at risk" and surfaced on the dashboard watch list.

Important Note:

The 50% threshold is a screening rule tuned on synthetic validation data. A controller can adjust it based on operational tolerance for false alarms versus missed delays, and it should be re-validated once trained on real corridor data.

Recommendation Engine

1

Risk Scoring

Every scheduled stop receives a delay-probability and expected-duration prediction.

2

Connection Buffer Check

For at-risk feeder trains, check whether the predicted delay breaches a downstream connection's transfer buffer.

3

Platform Capacity Check

If no tight connection exists, check whether the next station's platform capacity is likely to cause a knock-on wait.

4

Action Assignment

Each stop is assigned one action: Hold Connection, Reassign Platform, Hold Briefly, or Monitor.

WATCH-LIST REDUCTION SUMMARY

182 Scheduled Stops → 7 At-Risk Trains → 9 Recommended Actions

From the Python-validated reference design — the submitted .rmp performs delay-probability scoring only.

Results & Validation

86.39%**Accuracy**

On 4,914-row test set

0.760**AUC**

Positive class: delayed

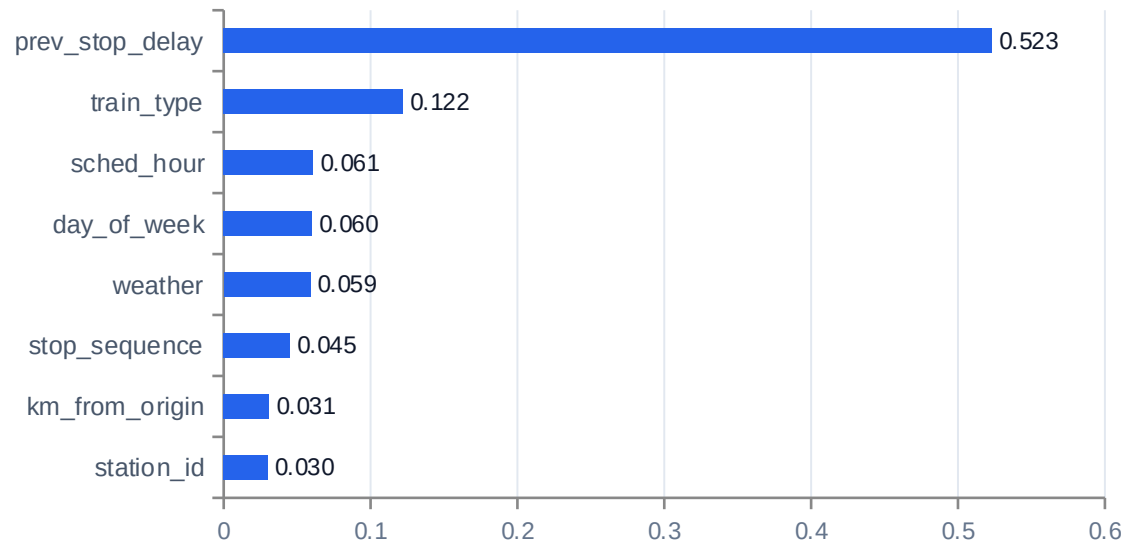
56.71%**Precision**

Delayed-class flags

42.61%**Recall**

Delayed-class flags

FEATURE IMPORTANCE — PYTHON REFERENCE MODEL



KEY INTERPRETATION

The trained Random Forest reached 86.39% accuracy and an AUC of 0.760 on held-out data — consistent with an independent Python validation (ROC-AUC 0.761) built on the same 13-feature set shown at left.

Recall (42.61%) trails precision (56.71%), meaning the model still misses more than half of true delays — a screening aid for controllers, not a fully automated call.

Conclusion & Future Enhancements



CONCLUSION

This project demonstrates that cascading railway delays are meaningfully predictable from schedule and upstream-delay data, and that predictions can be turned into concrete, prioritized dispatch recommendations rather than a raw risk score.

KEY OUTCOME

A 182-stop daily schedule is reduced to 7 flagged trains and 9 concrete recommendations

— turning a full day's operations into a short, actionable watch list for controllers, instead of requiring manual review of every service.

FUTURE ENHANCEMENTS

- Retrain on real operational delay data once available (e.g. published rail delay logs).
- Extend recommendations into a joint optimizer scheduling holds and platforms network-wide.
- Model downstream impact as a graph over the full connection network, not pairwise averages.
- Deploy the AI Studio model behind a live API for real-time, not daily-batch, predictions.

AI Studio Model



Domain Review



Real-Time API



Network Optimizer